Ashish Sukumar

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Professional Summary

Computer Vision and Deep Learning engineer pursuing M.S. in Robotics Engineering, seeking a PhD Autonomy Engineer Intern role focused on autonomous flight and 3D perception. Experienced building end-to-end deep learning systems for object detection, depth estimation, visual-inertial odometry, 3D scene reconstruction (NeRF, SfM), and sensor fusion using Python, C++, PyTorch, and OpenCV. Strong foundation in camera geometry, pose estimation, multi-view reconstruction, and motion planning. Proven ability to implement research papers from scratch, prototype rapidly, and push algorithms into working systems.

Education

Worcester Polytechnic Institute — **Massachusetts, USA**
M.S. Robotics Engineering — *2025–Present*

GPA: 4.00 / 4.00

Relevant Coursework: Computer Vision (RBE 549), Deep Learning for Robotics, Motion Planning, Image Processing, 3D Vision and Scene Reconstruction, Neural Networks and Pattern Recognition

SRM Institute of Science and Technology — **Chennai, India**
B.Tech Computer Science and Engineering — *2021–2025*

CGPA: 9.54 / 10

Technical Skills

Languages: Python (primary), C++

AI / ML Frameworks: PyTorch (primary), TensorFlow, Keras, CNNs, ResNet, Transformers, Spatial Transformer Networks

Computer Vision: OpenCV, Object Detection (YOLOv8), Depth Estimation, Image Segmentation, Feature Extraction, Optical Flow (RAFT), Homography Estimation, Camera Calibration, Epipolar Geometry

3D Vision / SLAM: NeRF, Structure from Motion (SfM), Point Cloud Processing, COLMAP, Pose Estimation, Bundle Adjustment, SLAM, Visual-Inertial Odometry (VIO), Stereo Vision, Kalman Filtering, Sensor Fusion

Robotics / Planning: ROS2, MoveIt, OMPL, Webots, Gazebo, Motion Planning, Dynamics and Controls

Tools: Linux, Git, Blender, Docker, CUDA

Experience

e-Yantra (IIT Bombay) — **Chennai, India**
Junior Project Technical Assistant — *Jun 2024–Feb 2025*

- Developed **computer vision** and **image processing** pipelines in **Python** and **OpenCV** for autonomous robot perception within competitive robotics environments.
- Built and tested robot simulation environments with autonomous task execution pipelines in **Webots Simulator**, integrating perception with decision-making logic for navigation and obstacle avoidance.
- Conducted **two technical workshops** on Embedded Systems and Robotics, formally recognised by Prof. Kavi Arya (Principal Investigator, e-Yantra, IIT Bombay). Produced structured technical documentation for national deployment.

Robocare Lab, Worcester Polytechnic Institute — **Worcester, MA**
Voluntary Research Assistant — *Oct 2025–Dec 2025*

- Contributed to research on **multimodal human-robot interaction (HRI)** using the SoftBank **Pepper Robot**, integrating speech, gesture, and visual feedback modalities via **ROS2**.
- Configured and tested robot **perception and feedback pipelines** using onboard cameras, microphones, and joint actuators for controlled lab experiments.

Projects

Deep Visual-Inertial Odometry for Autonomous Flight: Implemented a classical **Stereo Multi-State Constraint Kalman Filter (S-MSCKF)** pipeline fusing IMU and stereo camera data, achieving **0.12 m RMSE ATE** on EuRoC MAV benchmark after SE(3) alignment. Designed and trained three **deep learning** models in **PyTorch** (vision-only, IMU-only, visual-inertial fusion) for frame-to-frame relative **pose estimation** using multi-scale CNN encoders and bidirectional LSTMs with temporal attention. Built a synthetic **Blender**-based visual-inertial dataset (20 scenes, 10K poses each); applied **global trajectory optimization** with loop closure and smoothness constraints, reducing test ATE to **1.18 m** (39% improvement over dead-reckoning).

Einstein Vision – Autonomous Driving Scene Visualization: Built a modular **autonomous driving perception pipeline** processing dashcam video through **YOLOv8** object detection + ByteTrack multi-object tracking, **Depth Anything V2** monocular depth estimation, HybridNets lane detection, and **RAFT optical flow** for vehicle state estimation. Implemented ego-motion compensation using **Sampson distance** and **homography-based** verification to distinguish moving vs. parked vehicles; added **brake light detection** and **collision prediction**. Rendered full 3D scene reconstructions in **Blender** with color-coded vehicle states, procedural traffic signals, and pedestrian pose visualization (**YOLOv8-Pose**) across 13 real-world driving sequences.

Neural Radiance Fields (NeRF) – 3D Scene Reconstruction: Implemented a complete **NeRF** pipeline from scratch in **PyTorch** – positional encoding, hierarchical coarse-fine sampling, and volume rendering – achieving **27.42 dB PSNR / 0.9084 SSIM** on standard benchmarks. Extended to custom real-world datasets using **COLMAP**-based **Structure from Motion** for camera pose extraction. Debugged training instabilities and optimized data loading pipelines for real-world image sets.

Structure from Motion (SfM) Pipeline – Multi-View 3D Reconstruction: Built a full incremental **SfM** pipeline: **SIFT** feature matching, **RANSAC**-based Fundamental matrix estimation, Essential matrix decomposition, cheirality-checked triangulation (linear + nonlinear refinement), **PnP-RANSAC** camera registration, and **bundle adjustment**. Reconstructed **1,318 3D points** across 4 cameras from 5 real images – applicable to geometric world modeling for robotics and AR applications.

HomographyNet – Deep Learning Geometric Estimation: Implemented supervised and self-supervised **homography estimation** models in **TensorFlow** and **PyTorch**. Leveraged **Spatial Transformer Networks** to learn geometric transformations end-to-end from image pairs, benchmarking supervised vs. self-supervised approaches on synthetic warped datasets.

Publications

Ashish, S., Jeya, R., Rithish, R., Donipart, R., & Ganesh, K. (2025). *Fire Detection and Risk Prediction for Smart Safety: A Neural Network-Driven IoT Approach.* IJERCSE.

Ashish, S., Jeya, R., Rithish, R., Donipart, R., & Ganesh, K. (2025). *Fire Aware Smart-Bot with AI Responsive System.* ICIoT 2025.

Certifications & Achievements

- Red Hat Certified System Administrator (RHCSA)
- Oracle Cloud Infrastructure (OCI) Certified
- Excellence Award for Outstanding Student & Best Class Representative – SRMIST
- 2nd Place – Project Expo, SRMIST